Influence of Dynamical Decoupling Sequences with Finite-Width Pulses on Quantum Sensing for AC Magnetometry

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Dynamical decoupling sequences with multiple pulses can be considered to exhibit filter functions for the time evolution of a qubit superposition state. They contribute to the increase of coherence time and qubit-phase accumulation due to a time-varying field and can thus be used to achieve high-frequency-resolution spectroscopy. Such behaviors find useful application in highly sensitive detection based on qubits for various external fields, such as a magnetic field. Hence, decoupling sequences are indispensable tools for quantum sensing. In this study, we experimentally and theoretically investigate the effects of finite-width pulses in the sequences on ac magnetometry using nitrogen-vacancy centers in an isotopically controlled diamond. We reveal that the finite pulse widths cause a deviation of the optimum time to acquire the largest phase accumulation due to the sensing field from that expected by filter functions ignoring the pulse widths, even if the widths are considerably shorter than the time period of the sensing field. Moreover, we experimentally demonstrate that the deviation can be corrected by an appropriate time-frequency conversion. Our results provide a guideline for the detection of an ac field with an accurate frequency and linewidth in quantum sensing with multiple-pulse sequences.

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I. INTRODUCTION

The phase of a superposition state of a qubit is sensitive to various fields, such as a magnetic field. Recently, qubits have attracted considerable attention as highly sensitive sensors called “quantum sensors” [1,2], which can be used for a wide range of applications, such as magnetometry [3–6], thermometry [7–9], and electrometry [10,11]. Dynamical decoupling sequences with multiple pulses are broadly used as fundamental tools in not only quantum information processing but also quantum sensing. They can increase the coherence time of a quantum sensor [12], and thus its frequency resolution and sensitivity [13]. Moreover, the decoupling sequences possess filter functions for qubit-superposition phases, rendering the qubits sensitive to the fields with their time periods matched to that of the filters. Further, π pulses in the sequences reverse time evolutions of qubit states and modulate the filters, which enables arrangement of their properties. Such techniques have greatly advanced quantum sensing and have led to the realization of time-correlation spectroscopy [14–16] and measurements with ultrahigh-frequency resolution that is sufficient to detect chemical shifts via nuclear-magnetic-resonance signals [17–19]. For these applications, the amplitude of an alternating-current (ac) magnetic field is generally measured by ac-magnetometry techniques based on dynamical decoupling sequences. A qubit can obtain the maximum phase accumulation when the time period of an ac field matches that of the filter functions of multiple-pulse decoupling sequences. This phase-accumulation condition can enhance the quantum-sensor signal from a particular-frequency field and suppress the influence of noise caused by unwanted-frequency fields. Therefore, a qubit with multiple-pulse decoupling sequences can be regarded as a function of a lock-in amplifier in the quantum regime [20], which is used for detection of an ac magnetic field with a very small amplitude [3,5,21].

Characterization of the dynamical decoupling sequences is generally modeled and discussed with use of filter functions with infinitely narrow pulses [3,22,23]. In practice, however, the pulses have finite widths, creating the need to modify the filter functions. In this study, we propose a model of finite-pulse-width effects on ac magnetometry by using multiple-pulse sequences; in addition, we experimentally verify the model using nitrogen-vacancy (NV) centers in a diamond.

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II. MODEL OF FINITE-PULSE-WIDTH EFFECTS ON AC MAGNETOMETRY

A. Filter function with finite-width pulses

Dynamical decoupling sequences with multiple pulses have been well studied [22,23]. Their behavior is characterized by filter functions $F(t) = \pm 1$, where the sign changes whenever a π pulse is applied. If the pulse width is ignored, the filter function shows a square waveform, where the modulation timing can be controlled by π pulses. However, when a finite pulse width is considered, the filter function must be modified. Assuming that a qubit-control field generating π pulses is significantly stronger than the sensing field, the rotation axis of the qubit precession is parallel to the vector superposition of both fields, which is approximately in the control-field direction. Therefore, phase accumulation due to the sensing field cannot occur during the generation of π pulses, and the filter function can be represented as shown in Fig. 1, where N is the number of π pulses, τ_π is the pulse width, τ is the free precession time between two pulses, and total time $T = N(\tau + \tau_\pi)$. The center time of the jth π pulse is obtained by $c_j T/N$, $c_j = (2j-1)/2$.

In our theoretical model, the filter function with finite-width π pulses is given by

$$F(t) = \sum_{j=1}^{N} F_j(t), \tag{1}$$

$$F_j(t) = \begin{cases} (-1)^{j-1}, & (j-1)\dfrac{T}{N} \le t < c_j \dfrac{T}{N} - \dfrac{\tau_\pi}{2}, \\ (-1)^j, & c_j \dfrac{T}{N} + \dfrac{\tau_\pi}{2} \le t < j\dfrac{T}{N}, \\ 0, & \text{otherwise.} \end{cases} \tag{2}$$

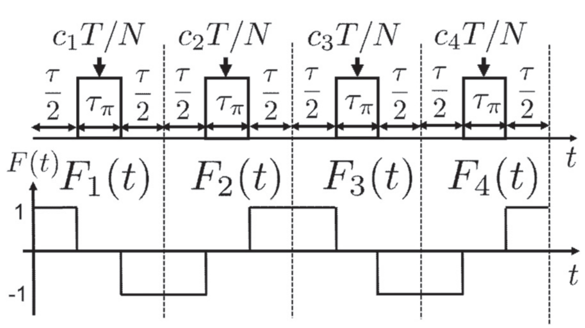

FIG. 1. A decoupling pulse sequence (top) and a filter function (bottom) with N finite-width pulses. τ_π is the pulse width, and τ is the free precession time between two pulses. The total time comprising the precession time and the width is $T = N(\tau + \tau_\pi)$. The center time of the jth π pulse is given by $c_j T/N$, $c_j = (2j-1)/2$. The sequence with $N = 4$ is shown as an example.

Through the Fourier transformation of the filter function, we have

$$\begin{aligned} F(\omega) &= \int_{-\infty}^{\infty} dt F(t) e^{-i\omega t} \\ &= \frac{1}{i\omega}\Bigg[1 + (-1)^{N+1} e^{-i\omega T} \\ &\quad + 2\sum_{j=1}^{N} (-1)^j e^{-i\omega c_j T/N} \cos\frac{\omega \tau_\pi}{2}\Bigg]. \end{aligned} \tag{3}$$

Michael *et al.* [24] reported a similar derivation for the field of quantum information by using an ion trap. To examine the finite-pulse-width effects on ac magnetometry, we calculate the Fourier component in the sequence with $N = 1$, as follows:

$$F(\omega) = \frac{i4e^{-i\omega[\tau(1+\alpha)]/2}}{\omega} \sin\left(\frac{\omega\tau}{4}\right) \sin\left[\frac{\omega\tau(1+2\alpha)}{4}\right], \tag{4}$$

where $\alpha = \tau_\pi/\tau$. Equation (4) corresponds to a filter function for a Hahn-echo sequence. For a sequence with even-N pulses, the Fourier component is given by

$$\begin{aligned} F(\omega) &= N\tau(1+\alpha)e^{-i\omega N\tau(1+\alpha)/2} \\ &\quad \times \left\{1 - \frac{\cos(\alpha\omega\tau/2)}{\cos[\omega\tau(1+\alpha)/2]}\right\} \frac{\sin[\omega N\tau(1+\alpha)/2]}{\omega N\tau(1+\alpha)/2}, \end{aligned} \tag{5}$$

which is a filter function for sequences of Carr-Purcell (CP) type [25].

B. Phase accumulation induced by an ac magnetic field

We assume a single-tone ac magnetic field given by

$$B(t) = B_{ac}\cos(\omega_{ac} t + \phi_{ac}), \tag{6}$$

where $\omega_{ac} = 2\pi f_{ac}$ (where f_{ac} is the field frequency), B_{ac} is the field strength, and ϕ_{ac} is the initial phase at the start of the decoupling sequences. A quantum sensor acquires phase accumulation given by

$$\begin{aligned} \Phi &= \int_{-\infty}^{\infty} dt F(t) \gamma_{NV} B(t) \\ &= \frac{\gamma_{NV} B_{ac}}{2}\left[e^{i\phi_{ac}} \int_{-\infty}^{\infty} dt F(t) e^{i\omega_{ac} t} \right. \\ &\quad \left. + e^{-i\phi_{ac}} \int_{-\infty}^{\infty} dt F(t) e^{-i\omega_{ac} t}\right] \\ &= \frac{\gamma_{NV} B_{ac}}{2}\left[e^{i\phi_{ac}} F^*(\omega_{ac}) + e^{-i\phi_{ac}} F(\omega_{ac})\right] \\ &= \gamma_{NV} B_{ac} \mathrm{Re}[e^{-i\phi_{ac}} F(\omega_{ac})], \end{aligned} \tag{7}$$

where γ_{NV} is the gyromagnetic ratio of the quantum sensor. In this study, we use NV centers in a diamond as quantum sensors; therefore, $\gamma_{NV}/2\pi = 28.03$ GHz/T. According to Eqs. (4) and (5), phase accumulation can be given as follows, on the basis of the Hahn-echo and the CP-type sequences:

$$\Phi_{\text{echo}} = \frac{4\gamma_{NV}B_{\text{ac}}}{\omega_{\text{ac}}} \sin\left[\frac{\omega_{\text{ac}}\tau(1+\alpha)}{2} + \phi_{\text{ac}}\right] \times \sin\left(\frac{\omega_{\text{ac}}\tau}{4}\right) \sin\left[\frac{\omega_{\text{ac}}\tau(1+2\alpha)}{4}\right], \quad (8)$$

$$\Phi_{\text{CP}} = \gamma_{NV}B_{\text{ac}} \cos\left[\frac{\omega_{\text{ac}}N\tau(1+\alpha)}{2} + \phi_{\text{ac}}\right] \times N\tau(1+\alpha)\left\{1 - \frac{\cos(\alpha\omega_{\text{ac}}\tau/2)}{\cos[\omega_{\text{ac}}\tau(1+\alpha)/2]}\right\} \times \frac{\sin[\omega_{\text{ac}}N\tau(1+\alpha)/2]}{\omega_{\text{ac}}N\tau(1+\alpha)/2}. \quad (9)$$

Commonly, a $\pi/2$ pulse is applied to a NV quantum sensor before the dynamical decoupling sequence, which creates the superposition state. Furthermore, an additional $\pi/2$ pulse is applied after the sequence, which enables optical detection of phase accumulation from NV centers. When both $\pi/2$ pulses are in phase, the ac-magnetometry signal is proportional to $\cos\Phi$. When the final $\pi/2$ pulse that is in quadrature with the other $\pi/2$ pulse is applied, the magnetometry signals become proportional to $\sin\Phi$.

III. EXPERIMENTS

A. Methods

We use an ensemble of NV centers (approximately 3×10^{14} cm^{-3}) as a quantum sensor. The NV centers are produced by a procedure reported in Ref. [26], with some modification. Briefly, a ^{12}C diamond film is prepared by microwave-plasma-assisted chemical vapor deposition from isotopically enriched $^{12}CH_4$ (purity greater than 99.999% for ^{12}C) and H_2 mixed gas. The thickness of the diamond film is 50 nm. We use a reactant gas with nitrogen-to-carbon ratio of 1.75% during the growth, and photolithography is used instead of electron-beam lithography. We evaluate the coherence time and relaxation time of the ensemble: $T_2 = 74.0 \pm 2.7$ μs and $T_1 = 7.04 \pm 0.27$ ms.

The ac-magnetometry measurements are performed with a homebuilt laser scanning microscope with an objective with a numerical aperture of 0.85 (LCPLFLN100xLCD, Olympus). Our microwave setup consists of a microwave source (FSW-0020 QuickSyn synthesizer, National Instruments) and a quadrature hybrid coupler (QH-0R714 quadrature hybrid coupler, Marki) with each output connected to a microwave switch (ZYSWA-2-500DR, Mini-Circuits) for generation of either in-phase or quadrature-phase microwave pulses. *On-off* timing of the microwave switches is controlled by transistor-transistor-logic (TTL) pulses generated by a TTL pulse generator (PulseBlasterESR-PRO-500, SpinCore), and thus the switches generate rectangular microwave pulses with a rise time and fall time of approximately 6 ns. After each switch, both output paths are combined and then amplified by a Mini-Circuits ZHL-16W-43+ amplifier, followed by a microwave antenna [27]. A commercial neodymium permanent magnet generates a static magnetic field of approximately 4 mT, with its direction parallel to a $\langle 111 \rangle$ axis of NV centers. An ac magnetic field is applied to the NV ensemble by a homemade coil connected to an arbitrary-function generator (AFG3252, Tektronix) and synchronized with dynamical decoupling sequences. We cancel common-mode noises in our measurements based on dynamical decoupling sequences by using a $3\pi/2$ pulse instead of a $-\pi/2$ pulse [12].

B. Results and discussion

To examine the effects of the finite pulse width, we perform ac magnetometry using a Hahn-echo sequence ($N = 1$); that is, $(\pi/2)_x$-$\tau/2$-$(\pi)_y$-$\tau/2$-$(\pi/2)_y$. A sensing field with $f_{\text{ac}} = 500$ kHz starts subsequent to the first $(\pi/2)_x$ pulse, and the field is synchronized to the Hahn-echo sequence. Figure 2(a) shows the ac-magnetometry results obtained with the Hahn-echo sequence with $\tau_\pi = 124$ ns. The solid line in Fig. 2(a) indicates the result fitted according to $\sin\Phi_{\text{echo}}$: $B_{\text{ac}} = 2.75 \pm 0.13$ μT and $\phi_{\text{ac}} = 91.4^\circ \pm 2.3^\circ$. Details of our fitting are explained in the Appendix. The largest phase accumulation is observed when the free precession time in the echo sequence is matched with the time period of the sensing field; that is, $\tau \approx (2k+1)T_{\text{ac}}$ $(k = 0, 1, 2, \ldots, T_{\text{ac}} = 1/f_{\text{ac}} = 2\ \mu\text{s})$. This feature is consistent with the expectations from the filter functions without consideration of finite-width pulses, as indicated by the dashed line in Fig. 2(a). By contrast, as the pulse width increases, the ac-magnetometry signals become increasingly different from those shown in Fig. 2(a). Figures 2(b) and 2(c) show the results obtained by ac magnetometry using the echo sequence with $\tau_\pi = 622$ and 1120 ns, which are equivalent to 5π and 9π. We observed finite-width-pulse-induced deviations of the phase-accumulated times from Fig. 2(a) and degradation of the ac-magnetometry signals (see Supplemental Material [28]). The solid lines in Figs. 2(b) and 2(c) are plotted with use of ac-magnetometry signals obtained with $\sin\Phi$ with $\Phi = \Phi_{\text{echo}}$ [Eq. (8)]. The solid lines in Figs. 2(b) and 2(c) obtained with the ac-field parameters evaluated from the results shown in Fig. 2(a) and each τ_π assigned to Eq. (8) show consistency with the experimental results.

Figure 3 shows the ac-magnetometry results based on a Carr-Purcell-Meiboom-Gill (CPMG) sequence [29] with

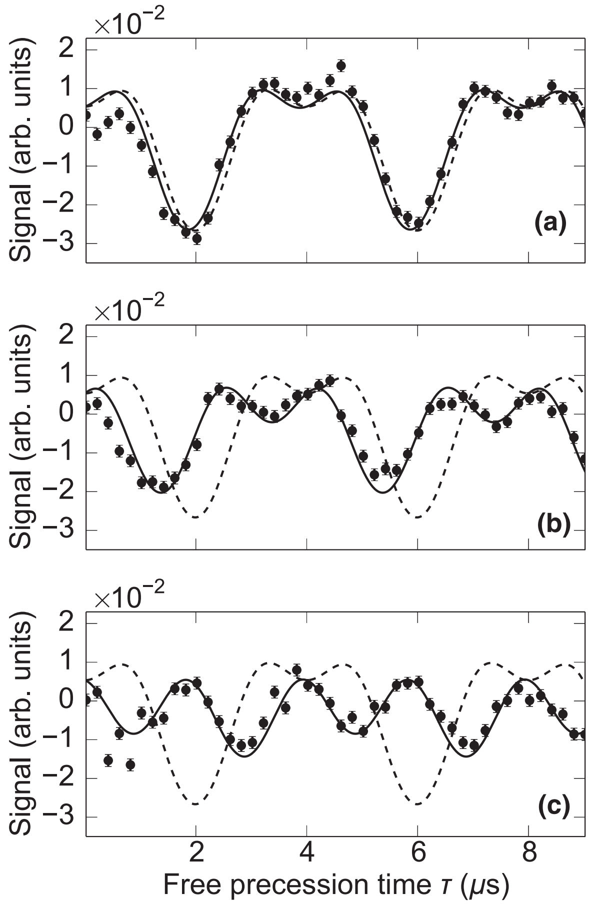

FIG. 2. Ac magnetometry using a Hahn-echo sequence. Error bars indicate the standard deviation of photon shot noise. The frequency of the sensing field is set at 500 kHz. Its strength and phase are evaluated by $\sin\Phi_{\mathrm{echo}}$ fitted to the results in the case of $\tau_\pi = 124$ ns, indicated by the solid line in (a). The detailed fitting procedure is explained in the Appendix. (b),(c) Ac magnetometry with $\tau_\pi = 622$ ns and $\tau_\pi = 1120$ ns, respectively. The solid lines in (b),(c) are obtained by $\sin\Phi_{\mathrm{echo}}$, where ac-field parameters are estimated from the results shown in (a) and each τ_π is assigned. The dashed line in (a)–(c) represents the theoretical plot without consideration of the pulse width, which is plotted by $\sin\Phi_{\mathrm{echo}}$, where $\tau_\pi = 0$ (i.e., $\alpha = 0$).

$N = 2$ (CPMG-2)—that is, $(\pi/2)_x$-$\tau/2$-$(\pi)_y$-τ-$(\pi)_y$-$\tau/2$-$(\pi/2)_y$—which is classified as the CP-type sequence. As described in the Appendix, $T_2(N = 2) = 94.7 \pm 3.5$ μs. The sensing-field strength and frequency are the same as those for the echo-based magnetometry, and we subtract 90° from the ac phase of our Hahn-echo measurements. According to $\sin\Phi_{\mathrm{CP}}$ given by Eq. (9), our fitting the experimental result with $\tau_\pi = 124$ ns yields $B_{\mathrm{ac}} = 2.42 \pm 0.09$ μT and $\phi_{\mathrm{ac}} = -2.53^\circ \pm 2.05^\circ$. We assign the evaluated ac-field parameters, $\tau_\pi = 374$, 622, and 872 ns, to Eq. (9), which reproduces our experimental results well.

Finally, ac-magnetometry results based on XY-series sequences with pulse errors suppressed [30] are obtained. Those sequences are classified as CP-type decoupling

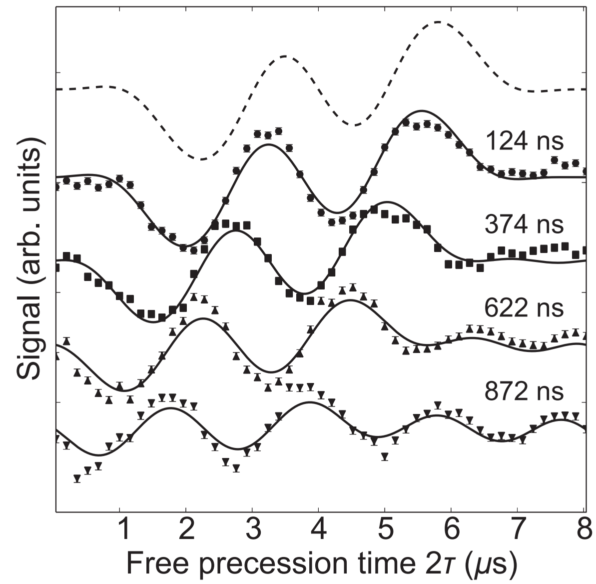

FIG. 3. Ac magnetometry using a CPMG-2 sequence. The frequency of the sensing field is set at 500 kHz. Its strength and phase are evaluated by $\sin\Phi_{\mathrm{CP}}$ fitted to the results with $\tau_\pi = 124$ ns. The fitting result is indicated by the solid line in the case of $\tau_\pi = 124$ ns. These ac-field parameters are used to obtain the solid lines plotted by $\sin\Phi_{\mathrm{CP}}$ for the results with $\tau_\pi = 374$, 622, and 872 ns, which are equivalent to 3π, 5π, and 7π pulses. The solid lines reproduce the results well. The dashed line represents the theoretical plot without consideration of the pulse width, which is plotted by $\sin\Phi_{\mathrm{CP}}$, where $\tau_\pi = 0$ (i.e., $\alpha = 0$). Signal degradation caused by the finite pulse width in the CP-type sequences is addressed in Supplemental Material [28].

sequences and are widely used for NV-based quantum sensing [19,23,31–38]. Figure 4(a) shows the ac-magnetometry results obtained with the XY8-5 sequence. The sensing-field frequency is set at 200 kHz, its estimated strength is 45.6 ± 2.9 nT, and $\phi_{\mathrm{ac}} = 181.2^\circ \pm 3.3^\circ$; τ_π is 126 ns. The dashed line in Fig. 4(a) represents the theoretical plot with infinitely narrow pulses ($\alpha = 0$). Although the optimum precession time for the phase accumulation is expected to be much longer than the widths, implying that $2\tau \sim 1/f_{\mathrm{ac}} = 5$ $\mu s \gg \tau_\pi$ and $\alpha \ll 1$, we observe the deviation of the peak from that indicated by $\sin\Phi_{\mathrm{CP}}$ with $\alpha = 0$, as seen in Fig. 4(a). If $\cos(\alpha\omega_{\mathrm{ac}}\tau/2) \sim 1$, Eq. (9) is given by

$$\Phi_{\mathrm{CP}} = \gamma_{\mathrm{NV}} B_{\mathrm{ac}} N\tau(1+\alpha) W_{\mathrm{CP}}(\omega_{\mathrm{ac}}, \phi_{\mathrm{ac}}), \tag{10}$$

where $W_{\mathrm{CP}}(\omega_{\mathrm{ac}}, \phi_{\mathrm{ac}})$ is given by

$$\begin{aligned} W_{\mathrm{CP}}(\omega_{\mathrm{ac}}, \phi_{\mathrm{ac}}) &\sim \frac{\sin[\omega_{\mathrm{ac}}N\tau(1+\alpha)/2]}{\omega_{\mathrm{ac}}N\tau(1+\alpha)/2}\left\{1 - \sec\left[\frac{\omega_{\mathrm{ac}}\tau(1+\alpha)}{2}\right]\right\} \\ &\quad\times \cos\left[\frac{\omega_{\mathrm{ac}}N\tau(1+\alpha)}{2} + \phi_{\mathrm{ac}}\right]. \end{aligned} \tag{11}$$

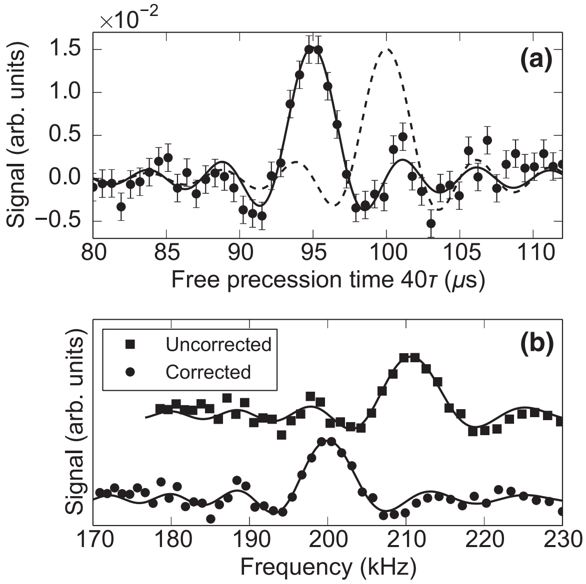

FIG. 4. (a) Ac magnetometry using an XY8-5 sequence ($N = 40$). XY-series sequences have the same filter functions as CP-type sequences. Error bars are obtained from the standard deviation of photon shot noise. The sensing-field frequency is set at 200 kHz. The solid line shows the fitting result according to $\sin \Phi_{\mathrm{CP}}$, whereas the dashed line in (a) indicates $\sin \Phi_{\mathrm{CP}}$ with $\alpha = 0$. (b) XY8-5 results plotted in a frequency range. Frequency conversion with $f = (2\tau)^{-1}$ causes a 10-kHz shift from the set frequency. By contrast, frequency conversion with $f = [2(\tau + \tau_\pi)]^{-1}$ corrects the deviation.

In our experimental conditions, this approximation is valid because $\alpha\omega_{\mathrm{ac}}\tau/2 \approx 0.025$ rad. $W_{\mathrm{CP}}(\omega_{\mathrm{ac}}, \phi_{\mathrm{ac}})$ in Eq. (11) is similar to a weighting function with infinitely narrow π pulses [see Eq. (66) in Ref. [1]], where τ is replaced with $\tau(1+\alpha) = \tau + \tau_\pi$. It implies that the frequency conversion using $f = [2(\tau + \tau_\pi)]^{-1}$ is available to correct the finite-pulse-width-induced deviation. Figure 4(b) shows the frequency plot of the XY8-based ac-magnetometry results. For the uncorrected experimental results plotted as a function of $f = (2\tau)^{-1}$ there is a 10-kHz shift of the main peak from the set frequency. In contrast, in the corrected conversion—that is, the data plotted as a function of $f = [2(\tau + \tau_\pi)]^{-1}$—the main peak appears at 200 kHz, which is consistent with the set frequency in this experiment. As shown in Fig. 4, the ac-magnetometry measurements using CP-type sequences with even pulses result in signals possessing a sinc function given by $\mathrm{sinc}[\omega N\tau(1+\alpha)/2]$ and their linewidths are approximately $1/T$, where $T = N\tau(1+\alpha)$. However, an uncorrected conversion plotted as a function of $f = (2\tau)^{-1}$ ignores α, which results in linewidths in a broader frequency range than that of the corrected-frequency plots. Therefore, finite-width pulses should be taken into account to evaluate accurate frequency and linewidth through ac magnetometry using dynamical decoupling sequences with multiple π pulses.

In this study, we used equidistant multiple-pulse sequences such as CPMG sequences for ac magnetometry. To improve frequency selectivity, filter functions designed by nonequidistant sequences have been proposed [39] and demonstrated [40]. The nonequidistant sequences can arrange the positions of main and side peaks of a filter function in a frequency range and moderate the strength of the peaks. We think that consideration of the pulse-width effect is necessary to design filter functions with their accurate frequencies by using the nonequidistant sequences. In particular, unless $\omega_{\mathrm{ac}}\tau_\pi \ll 1$ (i.e., in the case of measurements of high-frequency magnetic fields), we can deduce that the pulse width in dynamical decoupling sequences strongly affects the properties of filter functions with not only equidistant sequences but also with nonequidistant sequences.

Here we also comment on the influence of pulse width on shaped pulses. Pulse shaping has been well investigated to suppress the error of quantum control. Regarding ac magnetometry using a NV spin qubit, Jonathan *et al.* [41] reported the advantage of shaped pulses in dynamical decoupling sequences: pulses with their envelope shaped by a smooth function can increase the time and frequency resolutions of multiple-pulse decoupling sequences. They also demonstrated high-frequency ac magnetometry using the shaped pulses. We think that our filter-function model needs some modifications for ac magnetometry with such shaped pulses. As mentioned in Sec. II, our theoretical model is based on the assumption that a qubit-control field generating π pulses is significantly stronger than the ac magnetic field. However, this assumption is broken at the edges of pulses shaped by a smooth function. Therefore, we think that the discontinuous filter function [Eq. (2)] should be replaced by a smooth function.

IV. CONCLUSION

We study the effects of finite π-pulse widths on ac magnetometry theoretically and experimentally by using NV quantum sensors in an isotopically purified diamond film. Finite pulse widths comparable to the time period of the sensing field induce deviations of the optimum free precession times from the phase-accumulated times expected from the filter functions when finite-width-pulses are not taken into account. Furthermore, ac-magnetometry signals are degraded. Therefore, high-frequency field detection requires pulses as narrow as possible to evaluate the optimum time and increase the signal-to-noise ratio. Furthermore, long-time measurements with many π pulses may result in evident finite-width-pulse-induced deviations, even though the widths are much shorter than the time period of the sensing field. Such deviations cause failure in the estimation of the optimum free precession time for phase accumulation, resulting in the degradation of signals in ac magnetometry performed at fixed τ. Therefore, to estimate the optimum free precession time and detect ac signals with accurate frequency and linewidth, quantum sensing for ac magnetometry based on decoupling sequences with multiple pulses must take finite-width pulses into account.

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APPENDIX

In this study, a common-mode-rejection method is used to obtain the results [12]. Therefore, the ac-magnetometry signals obtained with the pulse sequences described in the main text are given by

$$S_Q = (-1)^{n_y+1}\frac{1-r}{1+r}\exp\left[-\left(\frac{N\tau}{T_2(N)}\right)^p\right]\sin\Phi, \quad \text{(A1)}$$

where r is the ratio of photon counts from the dark state to the bright state of the NV center, n_y is the number of $(\pi)_y$ pulses, $T_2(N)$ indicates the coherence time of N-pulse decoupling sequences, and Φ is the phase accumulation due to the sensing field, which is Φ_{echo} [Eq. (8)] in the measurements using the Hahn-echo sequences and Φ_{CP} [Eq. (9)] in the measurements using the CP-type sequences.

Using the Hahn-echo-sequence- and CPMG-sequence-based measurements without the sensing field, where the first and final $\pi/2$ pulses are in phase, we evaluate the parameters r, $T_2(N)$, and p, because the signals are given by

$$S_I = (-1)^{n_x+1}\frac{1-r}{1+r}\exp\left[-\left(\frac{N\tau}{T_2(N)}\right)^p\right], \quad \text{(A2)}$$

where n_x is the number of $(\pi)_x$ pulses. Figure 5 shows the Hahn-echo ($N=1$) measurement results, revealing $r = 0.895 \pm 0.001$, $T_2(N=1) = 74.0 \pm 2.7$ μs, and $p = 0.952 \pm 0.004$. In the CPMG-2 ($N=2$) measurement results shown in Fig. 5, $r = 0.892 \pm 0.001$, $T_2(N = 2) = 94.7 \pm 3.5$ μs, and $p = 1.11 \pm 0.06$. These evaluated parameters and Eq. (A1) are used for fitting the results described in the main text.

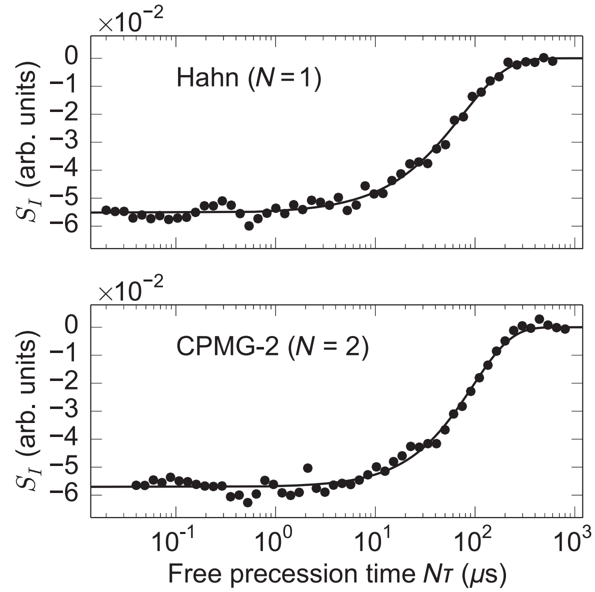

FIG. 5. Evaluation of coherence time of an ensemble of NV centers using the Hahn-echo ($N=1$) and CPMG-2 ($N=2$) sequences. We apply $(\pi)_y$ pulses for both sequences; that is, $n_x = 0$ in Eq. (A2).

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